

Causality in Oncology: Two Case Studies, Several Issues

Alessio Zanga^{1, 2, *} Alice Bernasconi^{1, 3} Peter J.F. Lucas⁴
Hanny Pijnenborg⁵ Casper Reijnen⁵ Marco Scutari⁶ Fabio Stella¹

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¹Models and Algorithms for Data and Text Mining (MADLab),
University of Milano - Bicocca, Milan, Italy

²Data Science and Advanced Analytics,
F. Hoffmann - La Roche Ltd, Basel, Switzerland

³Fondazione IRCCS Istituto Nazionale dei Tumori, Milan, Italy

⁴University of Twente, Enschede, The Netherlands

⁵RadboudUMC, Nijmegen, The Netherlands

⁶Istituto Dalle Molle di Studi sull'Intelligenza Artificiale (IDSIA),
Lugano, Switzerland

* Corresponding author: alessio.zanga@unimib.it

A (Very) Brief Introduction to Causality

Randomized Controlled Trials vs. Observational Studies

- Let X be a treatment, Y the outcome and Z a covariate.
- A Randomized Controlled Trial (RCT) is a prospective study where **treatment assignment** does not depend on any observed variable, i.e. it **is random**.
- An Observational Study (OBS) is a retrospective study where the treatment assignment mechanism **is not known**.

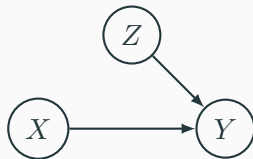


Figure 1: $\mathcal{G}_{\text{RCT}}$

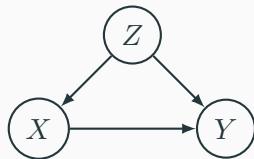


Figure 2: $\mathcal{G}_{\text{OBS}}$

Causal Effect vs. Confounding Bias

- The **causal effect** of a treatment X on an outcome Y is *the difference in mean outcome Y per unit of treatment X* [1]:

$$\mathcal{G}_{\text{RCT}} : \quad Y := \alpha \cdot X + \varepsilon$$

- When a covariate Z is a common cause of both X and Y , then the effect of Z is *entangled* with the effect of X :

$$\mathcal{G}_{\text{OBS}} : \quad Y := \alpha \cdot X + ? + \varepsilon$$

- Known as **confounding bias**.

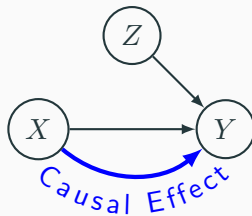


Figure 3: $\mathcal{G}_{\text{RCT}}$

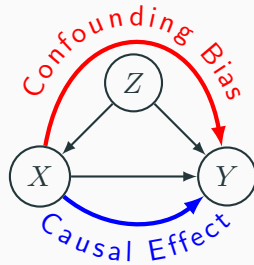


Figure 4: $\mathcal{G}_{\text{OBS}}$

Adjusting for Confounding Bias

- By looking at the **causal graph** it is possible to include the covariates that induce the confounding bias:

$$\mathcal{G}_{\text{OBS}} : \quad Y := \alpha \cdot X + \beta \cdot Z + \varepsilon$$

- Known as **covariate adjustment**.

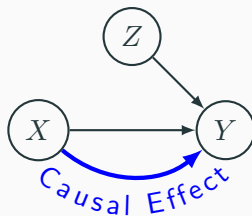


Figure 5: $\mathcal{G}_{\text{RCT}}$

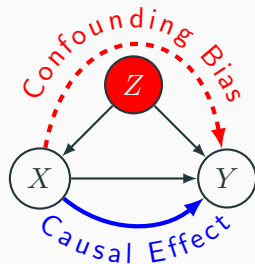


Figure 6: $\mathcal{G}_{\text{OBS}}$

Recovering the Causal Graph

- **Causal Discovery** [2] recovers the *true* graph $\mathcal{G}^*$ from data set $\mathcal{D}$ and prior knowledge $\mathcal{K}$.
- Following [3], we can write:

$$\underbrace{P(\mathcal{G} | \mathcal{D})}_{\text{Posterior}} \propto \underbrace{P(\mathcal{G})}_{\text{Prior}} \cdot \underbrace{P(\mathcal{D} | \mathcal{G})}_{\text{Likelihood}}$$

- Log-likelihood maximization:

$$\mathcal{G}^* = \arg \max_{\mathcal{G} \in \mathbb{G}} \log P(\mathcal{D} | \mathcal{G})$$

- In case of missing values, we can use the **Structural Expectation - Maximization (SEM)** [4] algorithm:
 1. Initialize $\mathcal{G}_0$,
 2. E Step - Fill $\mathcal{D}_{i+1}$ with $\mathcal{G}_i$,
 3. M Step - Learn $\mathcal{G}_{i+1}$ from $\mathcal{D}_{i+1}$.
 4. Repeat from (2) until convergence.
 5. Return $\mathcal{G}_{i+1}$.

From Theory to Application



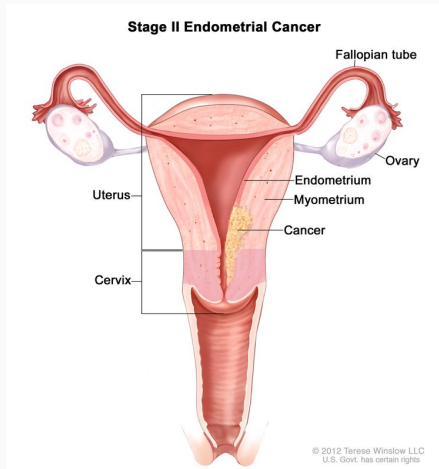
A General Overview

Case Study	LNM-EC		CVD-BC	
	#	Type	#	Type
Datasets	10	Hospitals	2	Pop / Clinic Based Cohorts
Samples	763	166 Max / 18 Min	1375	1036 PB / 339 CB
Treatments	2	Ch,Rd	8	[Ch,Rd,Tg,Hm] $\times$ [Neo,Adj]
Outcomes	4	Rec, Surv 1,3,5yr	1	Surv 5yr
Factors	1	Lymph Node Metastases	1	Cardiovascular Disease
Prior?	Yes	Partial Ord, Ltd Clinics	Yes	Partial Ord, Ext Clinics
Missing?	Yes	MNAR	Yes	MNAR
Overlapping?	Yes	Full	No	Partial
Selection?	No	-	Yes	Pop, Clinic

Table 1: A general overview of the two case studies.

Lymph Node Metastases (LNM) in Endometrial Cancer (EC) Patients

- **Endometrial cancer** from ten oncological clinics in the *European Network for Individualized Treatment of Endometrial Cancer* (ENITEC).
- Real-world data:
 - Small sample size.
 - High **missingness** ratio.
- Model-based approach:
 - **Prior knowledge** from clinicians.
 - **Missingness** mechanism.



[LNM-EC] Data

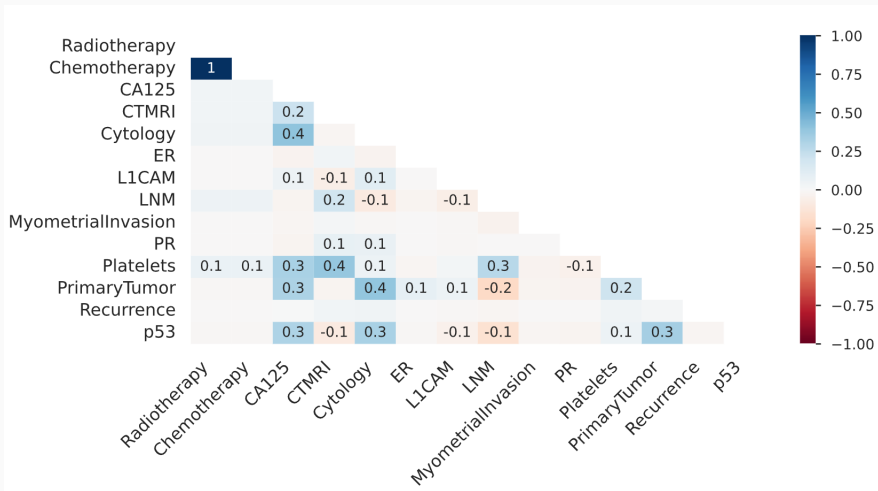


Figure 7: Nullity correlation matrix $\rho(\mathbf{R})$.

[LNM-EC] Methods - Part I - Definition of Missingness Graphs

A missingness graph [5] is a causal graph where the vertices are partitioned into *partially observed* variables $\mathbf{M}$, *proxy* variables $\mathbf{S}^*$ and *missing indicators* $\mathbf{R}$.

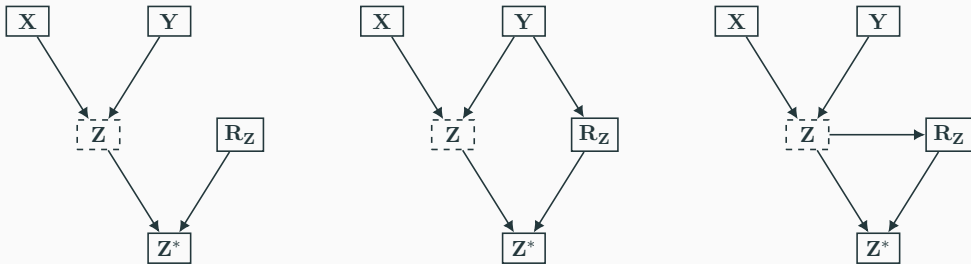


Figure 8: Missingness graphs modeling MCAR, MAR and MNAR respectively.

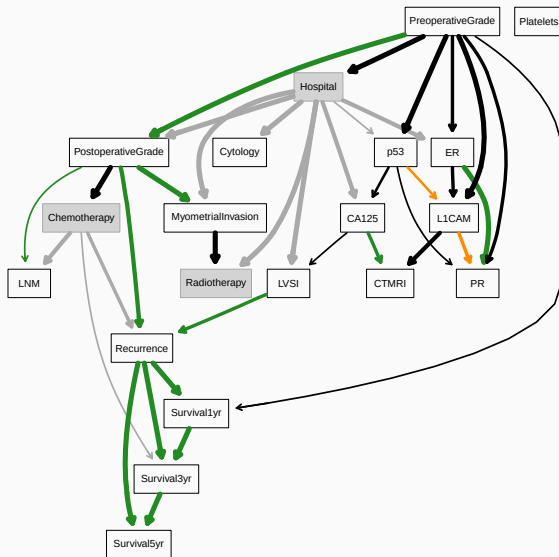
- MCAR implies $\mathbf{O} \cup \mathbf{U} \cup \mathbf{M} \perp\!\!\!\perp \mathbf{R}$,
i.e. missingness is random and independent from both fully- and partially-observed variables,
- MAR implies $\mathbf{U} \cup \mathbf{M} \perp\!\!\!\perp \mathbf{R} \mid \mathbf{O}$,
missingness is random conditioned on the fully-observed variables,
- MNAR if neither MCAR nor MAR.

- The joint distribution $P(\mathbf{V})$ is given by [5]:

$$P(\mathbf{V}) = \frac{P(\mathbf{R} = \mathbf{0}) \cdot P(\mathbf{V} \mid \mathbf{R} = \mathbf{0})}{\prod_i P(R_i = 0 \mid \boldsymbol{\Pi}_{\mathbf{R}_i}, \mathbf{R}_{\boldsymbol{\Pi}_{\mathbf{R}_i}} = \mathbf{0})}$$

[LNM-EC] Experiments - Part I - MCAR/MAR Assumption

- **Strength plot** of discovered graph [6] with the addition of the previously unobserved Hospital.
- **Green edges** are present both in reference and discovered graph, **orange edges** are reversed.
- **Gray nodes and edges** cannot be directly compared to reference graph.



[LNM-EC] Experiments - Part II - MNAR Assumption

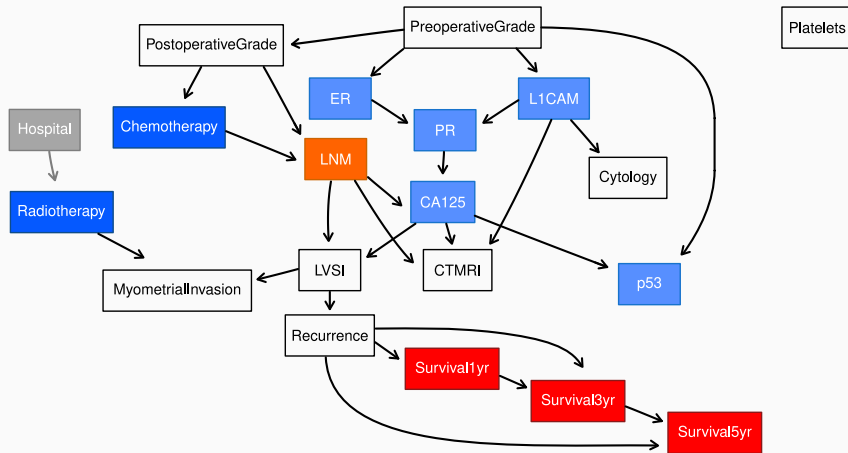


Figure 9: Causal graph recovered by HC-aIPW under MNAR [7].

Conclusions:

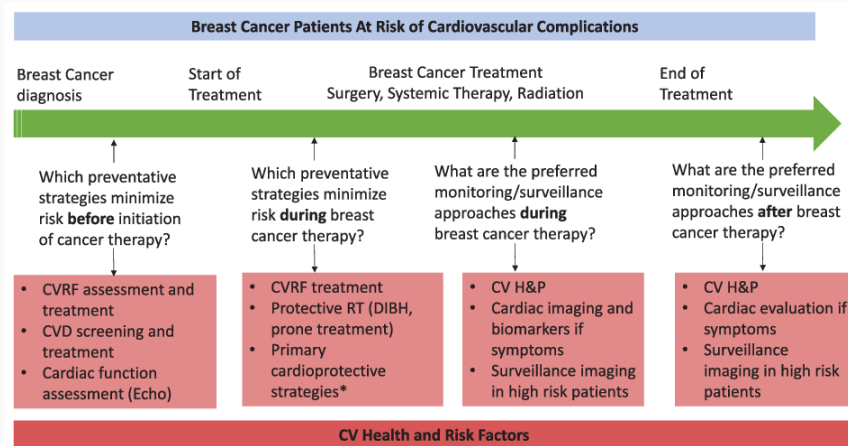
- Different **missingness mechanisms** lead to different models, even with the same data.
- **Imputation** may introduce more bias.
- **Prior knowledge** plays a crucial role.

Future Work:

- Check for **context-dependant missingness**.
- **Estimation** of the causal effect.
- Deal with **hidden variables**.

Cardiovascular Disease (CVD) in Adolescents and Young Adults with Breast Cancer (BC)

[CVD-BC] Introduction



Variables on	Population-based cohort	Clinical cohort
Patient	Birth date Incidence date Place of residence[Cat] Previous cancers[Y/N]	✓
Cancer	Histology[Cat], Topography[Cat], Grading[Cat]	✚ Stage[Cat], Recurrence[Y/N], Laterality[Cat] Receptor status[Cat]
Treatment	Type of Surgery[Cat], Chemotherapy[Y/N], Radiotherapy[Y/N], Hormone therapy[Y/N], Target therapy[Y/N]	✚ Chemo/target/homones pharmas[Cat] and doses[Cont], Radiotherapy site[Cat] and dose[Cont]
Cardiovascular risk factor	Hypertension[Y/N], Diabetes[Y/N], Dyslipidemia[Y/N]	✚ Body Mass Index[Cont], Systolic pression[Cont], Smoking habits[Cat]
Cardiovascular events	Event date, Description[Cat]	✘

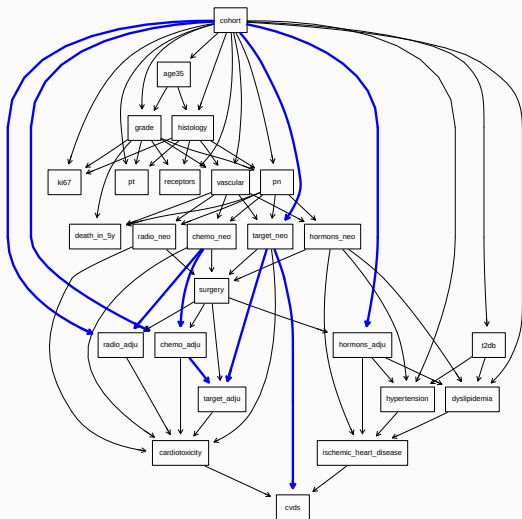
Definition 4.1 (Selection Diagram)

Let Π and Π^* be two different populations with a common underlying causal graph $\mathcal{G}$. A selection diagram $\mathcal{S}$ extends the causal graph $\mathcal{G}$ so that:

- $\mathbf{E}_{\mathcal{G}} \subset \mathbf{E}_{\mathcal{S}}$ that is, $\mathcal{G}$ is a sub-graph of $\mathcal{S}$;
- $\exists \mathbf{S} \neq \emptyset, \mathbf{S} \subset \mathbf{V}_{\mathcal{S}} \wedge \mathbf{S} \not\subset \mathbf{V}_{\mathcal{G}}$ where $(S_i \rightarrow V_i) \in \mathbf{E}_{\mathcal{S}}$ iff $f_i \neq f_i^*$, that is, the variables $\mathbf{S}$ point to the variables $\mathbf{V}_{\mathcal{G}}$ that differ in their value assignment f in Π and Π^* .

We combine a missingness graph and a selection diagram into a single causal graph using the SEM algorithm and clinical prior knowledge.

[CVD-BC] Experiments



Validation of the **SEM** results [8]:

- *Edges from [cohort] → selection mechanism [9];*
- *Edges directed from neoadjuvant [_neo] to adjuvant treatments [_adju] → surgical details not encoded in the [surgery] [10];*
- *Edges directed to cardiovascular diseases [cvds] → unstudied effects of oxidative stress and inflammation [11].*

Conclusions:

We proposed **the first causal network** for CVD risk in AYAs, capable of:

- Dealing with confounding bias,
- Combining data from different populations,
- Incorporating prior knowledge,
- Providing interpretable insights.

Future Work:

- Edges added by the causal discovery step provided **potential clinical insights**, especially for such an understudied population.
- We are working on extending the model to include **unmeasured confounders**.

Thank You for your Attention!

MADLab

The *Models and Algorithms for Data & Text Mining* Laboratory (MADLab) is a research lab at University of Milano - Bicocca focused on Causal Networks, Bayesian Networks and Continuous-Time Bayesian Networks applied to Healthcare and Medicine.

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